

Rojan Karki

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EDUCATION

Nepal College of Information Technology, Pokhara University

Bachelor of Software Engineering

3.64 CGPA

Lalitpur, Nepal

Aug 2022

- **Relevant coursework:** *Microprocessor and Computer Architecture, Data Structure & Algorithms, Artificial Intelligence and Neural Networks, Image Processing and Pattern Recognition, Distributed and Real Time Systems*

Little Angels' Secondary School

National Examination Board

77.2% (Distinction)

Lalitpur, Nepal

Aug 2017

PROFESSIONAL EXPERIENCE

Java Software Nepal

Software Engineer

Lalitpur, Nepal

May 2022 - Present

Resume Screening Tool

- Developed an in-house application to enhance the recruitment process.
- Designed and implemented a custom scoring algorithm to filter and rank qualified candidates from over 100 applications based on relevant experience and skills.
- Integrated a sentence transformer model (SBERT) to compute semantic textual similarity between job descriptions and resumes, significantly improving candidate-job match accuracy.

Tools: pdf2image, PyTesseract, SBERT, OpenCV

Bolsius Pimcore Implementation

- Developed a product information management system for a European client using Pimcore, replacing traditional ERP systems.
- Designed, implemented, and maintained an automated deployment process using GitLab CI/CD pipelines, resulting in a 90% reduction in manual workload.
- Collaborated with an international team of 10+ members to develop and implement features as per requirements.
- Architected and implemented an import process for over 30,000 products, each comprising more than 100 attributes.
- Engineered and implemented a highly efficient solution for importing and synchronizing assets, exceeding 100GB, between a remote server and a Pimcore application, using Samba.

TEACHING EXPERIENCE

CMP 440: Software Testing, Verification, Validation, and Quality Assurance

2025 FALL

- Undergraduate Course at Nepal College of Information Technology

UNIVERSITY PROJECTS

BERT based Conversational Chatbot

Sep 2022

- Developed a pluggable chatbot by fine-tuning BERT, providing automated responses to resolve customer queries regarding bank services

Room Finder And Recommender Using Collaborative Filtering

Oct 2021

- Developed web-based application for search and implemented content-based filtering for the recommendation system

Food Ordering System in Restaurant

Nov 2019

- Built a web application using PHP, JavaScript, and MySQL that can take orders, notify order completion, and automate billing.

RELEVANT PROJECTS [\[GitHub\]](#)

ML 101

- Implemented regression and neural models from scratch to explore core machine learning principles such as gradient descent, regularization, and forward propagation.

Brain Tumor Segmentation using U-Net architecture

- Implemented a 3D U-Net model for volumetric brain tumor segmentation on MRI data using subvolume sampling, intensity standardization, and dice loss optimization.

Machine Learning–Based Prediction of Automobile Prices Using Multivariate Regression Analysis

- Developed an automobile price prediction model using one-hot encoding and standardization for feature engineering, and implemented OLS and Stochastic Gradient Descent regression models for predictive analysis.

Definition-Guided Zero-Shot Classification of Student Sentiment on Mental Health Support

- Reproduced and extended the SMILE-College study, evaluating the free-tier Gemini 2.5 Flash on student responses using fine-grained sentiment classification.
- Benchmarked performance against GPT-3.5 and BERT baselines and analyzed key evaluation metrics.

SKILLS AND TECHNOLOGIES

Python, NLP, Pytorch, TensorFlow, Keras, Pandas, NumPy, PHP, Express.js, MySQL, Git, Bash

CERTIFICATIONS

Creative Problem Solving

June 2021

University of Minnesota ([Coursera](#))

Information Security: Context and Introduction

June 2021

University of London, Royal Holloway, University of London ([Coursera](#))

Machine Learning Specialization

June 2025

- *Supervised Machine Learning: Regression and Classification*
- *Advanced Learning Algorithms*
- *Unsupervised Learning, Recommenders, Reinforcement Learning*

DeepLearning.AI, Stanford University ([Coursera](#))

REFERENCES

- **Bhusan Thapa** (bhusan@ncit.edu.np)
Head of the Department of Software Engineering, NCIT